

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Conducted energy assessments and site surveys across industrial client facilities, collecting and analysing energy consumption data using **SQL** and **Python** to identify inefficiencies and develop energy saving opportunities aligned with **ISO 50001**, **EN 16247**, and **IS 399** standards.
- Supported preparation of technical reports, feasibility studies, engineering calculations, and design documentation for industrial decarbonisation projects; assisted with bills of quantities and cost estimation under senior engineer guidance, delivering **15+ project scopes** across HVAC, Steam, Chilled Water, LPHW, Compressed Air, and electrical distribution systems.
- Executed Measurement Verification (MV) activities post-implementation to track and validate project savings, using **Power BI** and **data modelling** techniques to build baseline energy dashboards and monitor **10+ industrial sites** for compliance with energy reduction targets.
- Attended client meetings and site visits, delivering technical communication and stakeholder engagement on industrial utility systems, change-control management, and post-implementation project triage across interconnected facilities with complex operational dependencies.
- Analysed industrial system data using **Python**, **SQL (PostgreSQL, MySQL, Oracle)**, and statistical methods to benchmark energy performance, identify improvement opportunities, and develop continuous improvement recommendations for manufacturing and food & beverage clients.
- Supported engineering team research, process development, and technical tooling improvements; contributed to internal training materials and documentation to strengthen Climeaction's technical delivery capability and professional development pathway toward chartership.

SKILLS

Energy & Sustainability Engineering – Industrial decarbonisation, energy assessments, site surveys, energy audits, baseline development, Measurement Verification (MV), ISO 50001, EN 16247, IS 399 standards, energy saving opportunity identification.

Industrial Systems & HVAC – HVAC systems, Steam systems, Chilled Water, LPHW (Low Pressure Hot Water), Compressed Air, Cooling Towers, electrical distribution, utility systems analysis.

Technical Analysis & Design – Feasibility studies, engineering calculations, design documentation, technical reports, bills of quantities, cost estimation, energy benchmarking, data collection.

Data Analysis & Business Intelligence – Power BI dashboards, data modelling, SQL (PostgreSQL, MySQL, Oracle), Python (pandas, NumPy, scikit-learn), statistical analysis, trend analysis, performance tracking.

Project & Stakeholder Management – Client meetings, site visits, technical communication, change-control management, UAT coordination, post-implementation support, cross-functional collaboration.

Software & Tools – Power BI, Python, SQL, Git, Jira, Linux, data visualization, engineering software, CAD concepts.

Certifications & Education – MSc Data Analytics (NCI Dublin), Bachelor of Engineering in Computer Science, IELTS Academic Band 7/9, Full Stack Java Spring Boot & Angular.

EDUCATION AND TRAINING

PROJECTS

Industrial Energy Baseline & Measurement Verification Platform (HCL Technologies)

- Designed and deployed an integrated energy data collection and baseline development platform supporting **12 multi-utility industrial sites** across manufacturing and food & beverage sectors, combining **SQL data pipelines** with **Python ETL automation** to standardize energy consumption data from HVAC, Steam, Compressed Air, and electrical systems.
- Built interactive **Power BI dashboards** with **DAX measures** and dynamic data models to track real-time energy KPIs, baseline deviations, and savings post-implementation, enabling **ISO 50001 / EN 16247 compliance reporting** and reducing manual reporting overhead by **70%**.
- Developed **Python-based Measurement Verification (MV) analytics** using pandas and scikit-learn to validate energy savings against baseline projections, achieving **±5% accuracy** in post-implementation performance tracking and supporting **€2.3M in verified annual energy cost reductions** across client portfolio.
- Created technical documentation, calculation libraries, and engineering templates for energy audits and feasibility studies; integrated cost estimation automation reducing proposal turnaround from **10 days to 3 days** and improving quote accuracy by **15%**.

Multi-Site Energy Systems Integration & Change Management (HCL Technologies)

- Supported cross-site energy systems integration across interconnected industrial facilities, coordinating change-control workflows and managing UAT activities for new HVAC and electrical distribution monitoring schemes affecting **8 critical production sites**.
- Conducted post-implementation site surveys and data validation across LPHW, Chilled Water, and Cooling Tower systems, resolving **12+ commissioning issues** and ensuring stable baseline establishment within **2 weeks of deployment**, supporting rapid Measurement Verification readiness.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)